

RAUNAK SINGH MATHARU

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EDUCATION

Northeastern University

Master of Science (MS) Information System

September 2022 – May 2024

Boston, MA

Mumbai University

Bachelor of Engineering (B.Tech) Electronics and Telecommunications

June 2018 – June 2022

Mumbai, India

SKILLS

Core Competencies	: Delta Lake Architecture, RFM Analysis, EDA Analysis, Normalization, Data Warehousing, CI/CD
Programming Language	: Python, SQL, PySpark, R, Advanced MS Excel, Excel Macros, T-SQL
Cloud Services	: AWS, GCP, AZURE, Snowflake, hivehadoop
Machine Learning	: Feature Engineering, Exploratory Data Analysis, Classification, Decision Tree, SVM, Random Forest
Tools/ Software's	: SQL (Postgres, MySQL, MS SQL Server), Docker, MS Visio, Oracle , Tableau, Power BI, MS Office
Other Skills	: advanced analytics, Business Intelligence, data models
Certification	: Microsoft Azure Data Engineer Associate , Azure Cloud Fundamentals , Tableau Desktop Specialist

WORK EXPERIENCE

EPIC

May 2023 – May 2024

Data Engineer

Lowell, MA

- Developed a pseudonymization ETL data pipeline that combined data from various sources using **AWS Lambda**, **PySpark**, and **S3** for secure storage, ensuring privacy for stakeholders, and implemented a streamlined CI/CD pipeline
- Contributed on creating and streamlining data pipelines from **S3** to **Snowflake** with the Beta and Product Data Team
- Automated data pipeline with **Athena**, **Snowflake**, **Airflow**, and **DBT** to aggregate tables reducing execution time by 20%
- Reconstructed SQL stored procedures to prevent SQL injection and triggers to automate auditing and data validation, improving performance by 20%
- Orchestrated and built batch processing ETL pipeline using, ensuring seamless data transformations and robust validation checks before data ingestion
- Independently enabled real-time analytics by building an **AWS Kinesis** based Change Data Capture (**CDC**) service with AWS Lambda to replicate **MongoDB** data to an **S3** bucket
- Worked with cross-functional teams to define data requirements and develop solutions in Patient Flow and Digital Patient Experience modules, enhancing productivity and outcomes by 30%
- Collaborated with senior leaders and stakeholders to troubleshoot any user acceptance issues after testing data integrity and ensured smooth deployment into production environment during each project cycle by using agile methodology

Magna Infotech

January 2021 – August 2022

Data Engineer

Mumbai, India

- Crafted and engineered ETL architecture with data pipelines using **Azure Data Factory**, **Azure Databricks** and **Logic Apps** with efficient loading of operational data into a data warehouse achieving accuracy rate of 99.9% for loaded data
- Led successful data migration to **Azure Synapse**, achieving 25% increase in performance and a 20% reduction in operational cost
- Implemented unit testing framework using python scripts and SQL scripts to ensure data quality, schema validation and accurate business logic transformations in **Databricks**
- Integrated comprehensive data cleansing framework, resulting in a 60% reduction in manual effort for cleaning data
- Eliminated manual efforts by Scripts and saved the team from receiving ad-hoc requests by Interactive Dashboard in **PowerBI** using filters, parameters, and **DAX** functions and customization according to business **KPIs**

PROJECTS

Crypto Insight Engine – Real-Time Crypto Data Extraction

- Engineered and deployed automated ETL pipelines using **Apache Airflow** in **Google Cloud Composer**, enabling data extraction using **REST APIs** every 10 minutes for cryptocurrencies to support real-time data analytics
- Enhanced data reliability through optimized workflows, enabling real-time analysis of 100+ different cryptocurrencies
- Collaborated on integrating **Airflow** with **Google BigQuery** for data integrity and to streamline incremental data loading

Query Cuisine – Restaurant Data Scraping

- Designed and established a real-time web scraping system using **BeautifulSoup**, with schema design for 8 tables and CI/CD pipelines for daily data extraction and managed workflows using **Airflow**
- Developed a user interface with **Streamlit** for interaction with **Snowflake** and created interactive visualizations with **Looker**

IMDB Movie Data Analysis

- Analyzed a movie rating dataset to recommend top movies, designing a data model with 14 dimensions and 4 fact tables with data ingestion from multiple sources, consolidating 50+ year dataset of movie roles in **SSMS** and maintaining **SCDs**
- Constructed a recommendation system using **LLM**, converting words to vectors and suggesting movies based on descriptions
- Created dynamic dashboards in **Tableau**, aligning SQL script output with visual data analysis to drive strategic insights

ADDITIONAL SKILLS

- Project management, business objectives, software development life cycle, teamwork, requirement analysis, strong interpersonal communication, and presentation skills, familiar working with Agile methodology, decision making, documentation cross-functional decision-making, data architecture, data models, data strategy, business objectives